WHEN BETTER GETS WORSE: IMPROVEMENT FIDELITY OF SELF-IMPROVEMENT OPERATORS IN ADAPTIVE WORLDS

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ABSTRACT

Self-improvement systems repeatedly replace an incumbent policy with a candidate that a verifier says is better. The hidden assumption is that this verifier-approved improvement transfers after deployment, even when the deployment changes the world. This implication can fail even when the verifier correctly evaluates every individual policy.

We therefore define *Improvement Fidelity* over the update transition $\tau = (\pi_t, \pi_{t+1})$, rather than over isolated policy values. The proxy improvement is $\Delta_V = J_V(\pi') - J_V(\pi)$ and the deployment-induced improvement is $\Delta_* = \mathcal{J}(\pi') - \mathcal{J}(\pi)$. An *improvement reversal* is $\Delta_V > 0$ with $\Delta_* < 0$: a verifier-approved replacement that becomes worse after the world responds.

PIVOT (Paired Interventional Validation of Optimization Transitions) is a budgeted intervention mechanism that preserves update decisions. Its primary rule, PIVOT-VOI, maintains a posterior over the differential correction $\Delta_* - \Delta_V$ and queries the transition with largest expected value of sample information per cost; the earlier transparent score is retained as PIVOT-H. It is not generic active learning and does not target global world-model error.

Across controlled mechanism evidence and a pinned public adaptive-environment pipeline, PIVOT exposes the transition-level estimand: powered E3b/E4b tests do not support a generic external CTI advantage, while a held-out strategic opponent family supports strategic reversal. The negative Binance stress test remains explicit. We do not claim causal market reversal, simulator ground truth, universal failure, or general multi-agent equilibrium; the contribution is a falsifiable evaluation criterion and a decision-theoretic implementation.

1 INTRODUCTION

Self-improvement is often described as a loop: generate a candidate, evaluate it, promote it, and repeat. But the central object optimized by the loop is not a policy in isolation; it is a *replacement operation*. A verifier may rank policies correctly while ranking improvements incorrectly, because deployment can change the transition and reward dynamics that define the next evaluation. The hidden implication is

$$\text{verifier says candidate is better} \quad \Longrightarrow \quad \text{deployment policy is better,} \tag{1}$$

and this implication need not hold even for a fixed, external, correctly implemented verifier.

This failure is distinct from self-authored verification, ordinary dataset overfitting, or an incorrect evaluation program. It is a transfer failure for the learning update itself: the proxy world evaluates an object that changes when the candidate is deployed.

We study the local transition

$$\tau_t = (\pi_t, \pi_{t+1}) \qquad \text{rather than only} \qquad J(\pi_t). \tag{2}$$

Let V be a cheap observer or replay world and let $*$ denote a deployment world that may respond to the policy. The proxy and deployment improvements are $\Delta_V = J_V(\pi') - J_V(\pi)$ and $\Delta_* =$

$\mathcal{J}(\pi') - \mathcal{J}(\pi)$. An improvement reversal is

$$\Delta_V > 0 \quad \wedge \quad \Delta_* < 0. \tag{3}$$

This definition makes no assumption about whether the proxy is learned, hand-coded, or externally audited. It asks whether the object consumed by the self-improvement loop transfers.

An improvement operator $\mathcal{A}$ induces a transition distribution $Q_{\mathcal{A}}$ over the replacements it proposes. We therefore define operator-relative Improvement Fidelity for any loss L by

$$\mathrm{IF}(V, \mathcal{A}; L) = \mathbb{E}_{(\pi,\pi')\sim Q_{\mathcal{A}}}\left[L(\Delta_V, \Delta_*)\right]. \tag{4}$$

The absolute-delta choice $L(a, b) = |a-b|$ is operator-conditioned IDE, while the sign-error choice is $1 - \mathrm{ISC}(Q_{\mathcal{A}})$ after excluding ties. IRR is the corresponding conditional failure probability. Thus the relevant population is induced by the self-improvement operator, rather than the uniform population of all policy pairs.

Our contributions are deliberately limited to three points:

1. **Contribution 1 — Principle and theory.** We introduce operator-relative *Improvement Fidelity*, prove Global Fidelity Blindness and an Operator Shift Bound, and retain the local response–footprint bound.
2. **Contribution 2 — Algorithm.** We introduce **PIVOT-VOI**, a paired posterior differential evaluator with EVSI-per-cost acquisition, adaptive stopping, and an explicit best-update sample bound; PIVOT-H is a transparent heuristic baseline.
3. **Contribution 3 — Evidence contract.** We provide controlled operator-shift and response-mechanism evidence plus a pinned external adaptive/strategic pipeline. Development underpowering and the negative finance boundary are reported as scientific outcomes, not hidden.

2 Related Work

Performative and strategic response. Performative prediction makes deployment part of the data-generating process Perdomo et al. (2020); Mandal et al. (2022), and recent work derives policy gradients or learning dynamics for performative RL and performative Markov games Basu et al. (2025); Sahitaj et al. (2025); Wang et al. (2025); Piliouras & Yu (2022). These papers study stable policies, equilibria, or convergence. They do not ask whether a particular replacement $\pi \to \pi'$ is still an improvement after the response is realized. **Gap:** PIVOT treats the update, not the fixed point, as the estimand.

World-model and simulator fidelity. WorldGym and policy-aware simulator learning evaluate policy values, rankings, or simulator objectives Quevedo et al. (2025); Dann et al. (2026); Hansen et al. (2024). Recent market simulators improve the realism of endogenous order flow and liquidity through interactive LOB feedback or generative state–event models Noble et al. (2026); Zhang et al. (2026); Kawawa-Beaudan et al. (2026). These advances still report policy-level fidelity or trajectory realism. **Gap:** they do not measure the sign or magnitude of the differential $J(\pi') - J(\pi)$ under a policy-induced world, so good global ranking can coexist with wrong local update ordering.

Self-improvement and verification. EVOQUANT, EvoPolicyGym, Darwin Godel Machine, and Evo-Harness make iterative policy or harness evolution executable Mao et al. (2026); Wang et al. (2026); Zhang et al. (2025); Wei et al. (2026); FinEvo-Bench extends evaluation to longitudinal financial workflows Deng et al. (2026). ICLR benchmarks such as AgentBench, WebArena, and SWE-bench make agent evaluation executable across interactive tasks Liu et al. (2024); Zhou et al. (2024); Jimenez et al. (2024). SEAL identifies the verifier–deployment gap when an agent authors its own tests, while recent stress tests expose variance, task-order, and execution-interface failures Guo et al. (2026); Ye et al. (2026); Li & Zhu (2026); Kulkarni et al. (2026). Those works either protect acceptance with a sealed oracle or benchmark the resulting agent. **Gap:** even a fixed external verifier can be correct about the wrong world; PIVOT validates transfer of the proposed update itself.

Decision-preserving queries and finance testbeds. Active learning and value-of-information provide the ancestry for allocating expensive queries Settles (2009); Lindley (1956); paired rollouts additionally exploit shared exogenous randomness. PIVOT changes the target from global prediction error to update-selection regret. ABIDES-MARL, FinEvo, and the NeurIPS FinRL-Meta benchmark, together with recent execution and market-impact simulators, supply increasingly endogenous finance testbeds Cheridito et al. (2025); Zou et al. (2026); Liu et al. (2022); Byrd et al. (2019); Cont et al. (2010); Nevmyvaka et al. (2006). **Gap:** these environments can reveal response mechanisms, but none defines improvement consistency or a budgeted transition-level validator that separates observer, actor, and strategic reversals.

3 PROBLEM FORMULATION

Policy-induced worlds. Deployment can alter transitions, rewards, liquidity, or other agents' behavior. We write this response as $\mathcal{M}[\pi]$ and define the deployment value

$$\mathcal{J}(\pi) = J(\pi; \mathcal{M}[\pi]). \tag{5}$$

For exposition, three layers separate mechanisms rather than assert that they are always identifiable:

$$\Delta_{\text{direct}} = J(\pi'; \mathcal{M}[\pi]) - J(\pi; \mathcal{M}[\pi]), \tag{6}$$

$$\Delta_{\text{actor}} = J(\pi'; \mathcal{M}[\pi']) - J(\pi; \mathcal{M}[\pi]), \tag{7}$$

$$\Delta_{\text{strategic}} = J_i(\pi', BR_{-i}(\pi')) - J_i(\pi, BR_{-i}(\pi)). \tag{8}$$

The first ignores response, the second includes the focal agent's mechanical footprint, and the third includes adaptation by competitors. In experiments we report these as separate worlds and do not label a learned or observational proxy as ground truth.

Improvement fidelity metrics. For n transitions, IDE is $n^{-1} \sum_k |(\Delta_V)_k - (\Delta_*)_k|$. ISC is the fraction of non-tied transitions whose signs agree. IRR is $P(\Delta_* < 0 \mid \Delta_V > 0)$, while SIRR is $P(\Delta_{\text{strategic}} < 0 \mid \Delta_{\text{actor}} > 0)$. When the rows are draws from $Q_{\mathcal{A}}$, IDE and $1 - \text{ISC}$ are the two empirical losses in $\text{IF}(V, \mathcal{A}; L)$ above; the operator label is therefore part of the estimand, not merely metadata. For a candidate set C_t, the update-selection regret is

$$\text{ISR}_t = \max_{j \in C_t} (\Delta_*)_{t,j} - (\Delta_*)_{t,\hat{j}}, \tag{9}$$

where $\hat{j}$ is selected by the proxy or corrected estimate. CTI sums the true improvements of promoted updates over a trajectory. We call this metric the cumulative true improvement (CTI). All estimates below retain query counts, seeds, and confidence intervals in the artifact snapshot.

4 PIVOT

PIVOT stands for Paired Interventional Validation of Optimization Transitions. Given an incumbent and K candidates, it first computes cheap proxy deltas and pre-query features. The features include proxy delta, response strength, competition strength, candidate index, and an update footprint. The generic footprint records policy-distance and action/support shifts; the finance footprint records size, participation, urgency, holding, turnover, liquidity consumption, and spread crossing.

Paired differential correction. For a queried transition, incumbent and candidate are evaluated on the same initial state, exogenous path, seed, and opponent initialization. PIVOT fits

$$g_\theta(z_\Delta) \approx \Delta_* - \Delta_V, \qquad \widehat{\Delta}_* = \Delta_V + g_\theta(z_\Delta). \tag{10}$$

Pairing subtracts common-mode noise before aggregation. The unpaired ablation uses disjoint context seeds and the standard independent-sample standard error.

Active query rule. Let D denote current paired evidence and let the posterior over true deltas define expected simple regret $\mathcal{R}(D) = \mathbb{E}[\max_j \Delta_j^* - \Delta_{a(D)}^* \mid D]$, where $a(D) = \arg\max_j \mathbb{E}[\Delta_j^* \mid D]$. A hypothetical HF observation Y_j induces

$$\text{EVSI}_j = \mathcal{R}(D) - \mathbb{E}_{Y_j|D}[\mathcal{R}(D \cup \{Y_j\})], \qquad A_j = \frac{\text{EVSI}_j}{c_j}, \tag{11}$$

and PIVOT-VOI queries $\arg\max_j A_j$. It stops when the posterior selection probability is at least $1 - \delta$ or $\max_j A_j < \eta$. The earlier transparent score $q_j = u_j(0.5 + r_j)(1 + o_j)/c_j$ is PIVOT-H, a baseline rather than the primary claim. Queried values replace predictions; unqueried candidates retain proxy/model estimates.

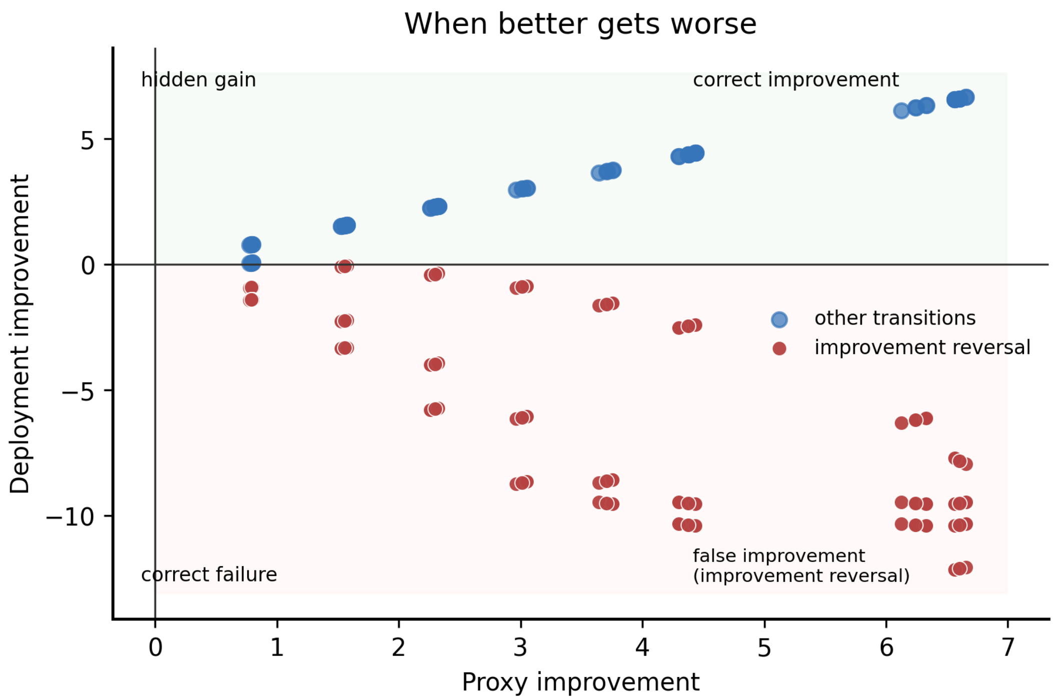

Figure 1: Controlled proxy versus deployment improvement. The upper-right region is correct improvement, the upper-left region is a hidden gain, the lower-left region is correct failure, and the lower-right region is the highlighted false improvement (improvement reversal). Each point is a transition in the frozen response–footprint grid.

Why Transition Validation Differs from Active Learning. Traditional active learning asks which label will reduce a global prediction loss for y. PIVOT asks which paired intervention can change the ordering or sign of an update. Its target is the differential correction $\Delta_* - \Delta_V$, and its loss is update-selection regret rather than global world-model error. In short, **PIVOT is not designed to minimize global world-model error; it targets the smallest intervention budget required to preserve optimization decisions.** This distinction is operational: an uncertain transition whose high-fidelity value cannot change the selected candidate has lower priority than an otherwise similar transition near the selection margin.

5 Theory

We state four results that motivate the differential estimand. They are not claims about a particular simulator architecture.

Proposition 1 (Global fidelity is sufficient)**.** *Suppose* $\sup_\pi |J_V(\pi) - J_*(\pi)| \le \varepsilon$*. Then every transition satisfies* $|\Delta_V - \Delta_*| \le 2\varepsilon$*. In particular, the sign is preserved whenever* $|\Delta_*| > 2\varepsilon$*.*

Proof. By adding and subtracting the two policy values, $\Delta_V - \Delta_* = [J_V(\pi') - J_*(\pi')] - [J_V(\pi) - J_*(\pi)]$. The triangle inequality gives the bound. If $|\Delta_*| > 2\varepsilon$, the perturbation cannot cross zero. $\square$

Proposition 2 (Global Fidelity Blindness)**.** *For every* $\varepsilon > 0$*, there is a finite policy family, a deployment value function* J_**, a verifier* V*, a policy distribution* P*, and an update operator* $\mathcal{A}$ *such that*

$$\mathrm{MAE}_P(J_V, J_*) < \varepsilon, \qquad \mathrm{Spearman}(J_V, J_*) > 1 - \varepsilon,$$

while, for the transition law $Q_\mathcal{A}$*,*

$$\mathrm{IRR}(Q_\mathcal{A}) = 1, \qquad \mathrm{ISC}(Q_\mathcal{A}) = 0.$$

Proof. Take n policies with uniform P and $J_*(i) = i/(n-1)$, and let V swap the values of one adjacent pair. Then $\mathrm{MAE}_P = 2/[n(n-1)]$ and the only nonzero rank displacements are $+1$ and

-1, so $1 - \text{Spearman} = 12/[n(n^2-1)]$. Choose n large enough that both quantities are below ε. Let $\mathcal{A}$ sample only the swapped directed transition (so $Q_{\mathcal{A}}$ has unit mass there) and propose the lower-valued true policy as a replacement for the higher-valued incumbent. The proxy delta is positive and the deployment delta is negative on every sampled transition, giving $\mathrm{IRR}(Q_{\mathcal{A}}) = 1$ and $\mathrm{ISC}(Q_{\mathcal{A}}) = 0$. ☐

Proposition 3 (Operator Shift Bound)**.** *Let P be the calibration law over transitions and let $Q_{\mathcal{A}} \ll P$ have density ratio $w = dQ_{\mathcal{A}}/dP$. For a non-negative transition loss $\ell(\tau) = L(\Delta_V(\tau), \Delta_*(\tau))$,*

$$\mathrm{IF}(V, \mathcal{A}; L) = \mathbb{E}_P[w(\tau)\ell(\tau)] \leq \sqrt{\mathbb{E}_P[\ell(\tau)^2]}\sqrt{1 + \chi^2(Q_{\mathcal{A}} \| P)}. \tag{12}$$

Proof. Absolute continuity gives the first equality. Cauchy–Schwarz bounds the expectation by $\sqrt{\mathbb{E}_P[\ell^2]}\sqrt{\mathbb{E}_P[w^2]}$. Since $\mathbb{E}_P[w^2] = 1 + \chi^2(Q_{\mathcal{A}} \| P)$, the result follows. ☐

Proposition 4 (Response–footprint bound)**.** *Let $d(\pi, \pi')$ be an update footprint and define the direct comparison in the incumbent response world, $\Delta_{\text{direct}} = J(\pi'; \mathcal{M}[\pi]) - J(\pi; \mathcal{M}[\pi])$. Assume the environment response satisfies $D(\mathcal{M}[\pi'], \mathcal{M}[\pi]) \leq L_M d(\pi, \pi')$ and the value functional is L_J-Lipschitz in its world argument. Then*

$$|\Delta_{\text{actor}} - \Delta_{\text{direct}}| \leq L_J L_M d(\pi, \pi'). \tag{13}$$

Proof. The incumbent term cancels because both comparisons use $\mathcal{M}[\pi]$ as the reference world. The remaining candidate term is bounded by $L_J D(\mathcal{M}[\pi'], \mathcal{M}[\pi]) \leq L_J L_M d(\pi, \pi')$. The result is local; it does not assert that strategic best responses are globally Lipschitz. ☐

Proposition 5 (Decision Preservation Under Differential Error)**.** *Consider two candidates i and j evaluated from the same incumbent, and suppose the true improvement margin is $m = \Delta_{*,i} - \Delta_{*,j} > 0$. If estimates satisfy $|\widehat{\Delta}_i - \Delta_{*,i}| < m/2$ and $|\widehat{\Delta}_j - \Delta_{*,j}| < m/2$, then $\widehat{\Delta}_i > \widehat{\Delta}_j$ and the selected update is unchanged.*

Proof. The estimated margin obeys $\widehat{\Delta}_i - \widehat{\Delta}_j = m + (\widehat{\Delta}_i - \Delta_{*,i}) - (\widehat{\Delta}_j - \Delta_{*,j})$. The two error terms have absolute sum strictly less than m, so the estimated margin remains positive. This is a local decision guarantee; it does not claim that PIVOT is globally optimal or that all candidates satisfy the margin condition. ☐

Proposition 6 (Finite-Sample Best-Update Identification)**.** *Suppose paired rollout differences are σ-sub-Gaussian and the best candidate has true margin at least $m > 0$ over each of $K-1$ competitors. With n paired rollouts per candidate, the probability of selecting a wrong update is at most*

$$(K-1)\exp\left(-\frac{nm^2}{4\sigma^2}\right). \tag{14}$$

Thus $n \geq 4\sigma^2 m^{-2} \log((K-1)/\delta)$ suffices for error at most δ (and the implementation uses the conservative K in its registry).

Proof. For each competitor, a wrong ordering requires the difference of two sample-mean errors to exceed m. The sub-Gaussian tail bound gives the stated pairwise probability; a union bound over $K-1$ competitors completes the proof. ☐

The first proposition explains why global value accuracy is sufficient but unnecessarily strong: a policy-independent offset can be arbitrarily large without changing any improvement difference. Global Fidelity Blindness shows that the converse fails when the update law concentrates on a locally swapped transition. The Operator Shift Bound quantifies the bridge: local IF can grow with the chi-square distance between the calibration law and $Q_{\mathcal{A}}$. The response–footprint result motivates targeted fidelity, while the final proposition connects differential error to the selection decision PIVOT must preserve. In strategic environments, small policy changes can still cause large competitor responses, so smoothness remains a diagnostic assumption.

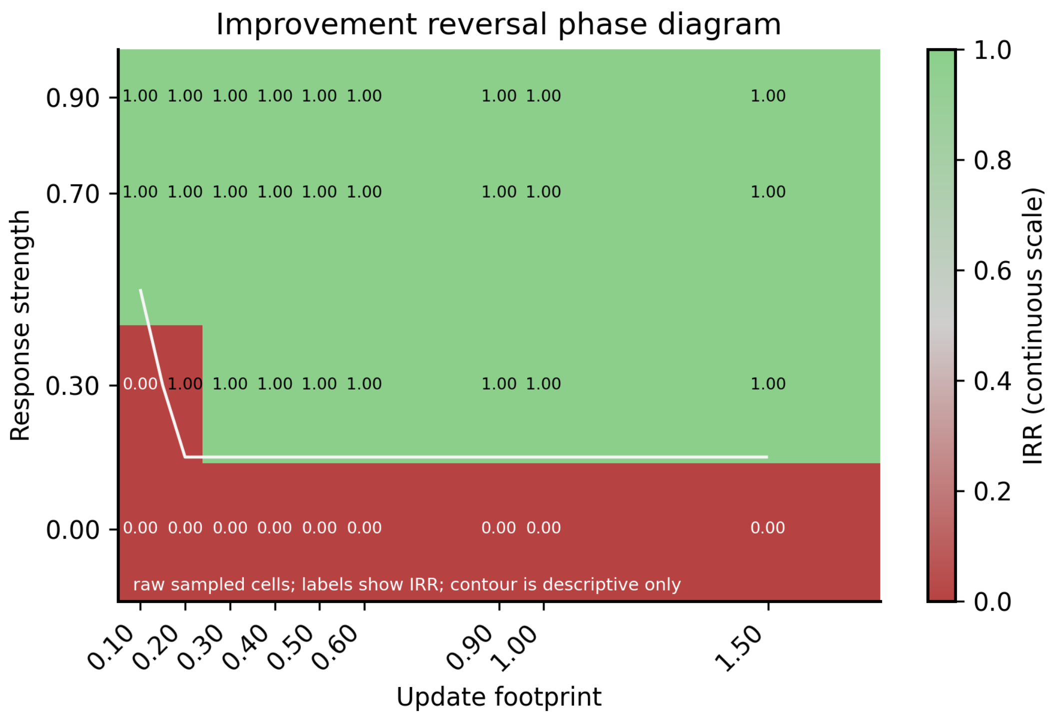

Figure 2: raw sampled reversal-rate cells over response strength and update footprint. Cell labels show observed IRR; the descriptive contour is not an interpolated empirical phase boundary.

6 Controlled Experiments

Setup. The controlled world has finite horizon 12, bounded rewards, Gaussian exogenous noise, and a tunable response strength. Candidate policies change one intensity component. P2 uses three independent registered run groups and a response grid $\{0, 0.3, 0.7, 0.9\}$, footprint scales $\{0.1, 0.2, 0.3, 0.5\}$, and optimization strengths $\{0.5, 1, 1.5\}$. All primary comparisons use disjoint registered seeds and paired high-fidelity transitions.

We also run the registered V6 analytic checks in the supplementary artifact. The Global Fidelity Blindness grid contains 105 rows: all three targets $\varepsilon \in \{10^{-1}, 10^{-2}, 10^{-3}\}$ are covered, the minimum Spearman correlation is 0.9971, and the minimum operator-conditioned IRR is 1.0. The response–footprint grid contains 180 rows and satisfies the stated bound on every row, with maximum observed/bound ratio $1 + 3.3 \times 10^{-15}$. These values are exact-construction checks and are kept separate from the controlled adaptive-world and finance evidence.

Operator shift diagnostic. The registered E2 population contains 63 paired transitions. We vary the temperature of a softmax improvement operator from 0.25 to 4, while the calibration population and verifier remain fixed. Chi-square shift rises from 0.0572 to 1.4466 and absolute operator-conditioned IF rises from 10.8833 to 15.5595; all rows satisfy the analytic bound. This supports the predicted operator-shift mechanism in the controlled world, not a claim about external adaptive environments.

The resulting decision-preserving data flow is summarized in fig. 3. The diagram treats a policy replacement as the unit of validation: cheap proxy scores screen the candidate set, while paired high-fidelity interventions are reserved for transitions that can change the selection decision.

Improvement reversal. Across 240 transitions per registered run, overall IRR is 0.7292 and the high-response IRR is 0.9722; low-response IRR is zero. The phase diagram in fig. 2 shows the intended response–footprint interaction. These results establish that a correct observer verifier can approve harmful updates in a world with policy-induced response.

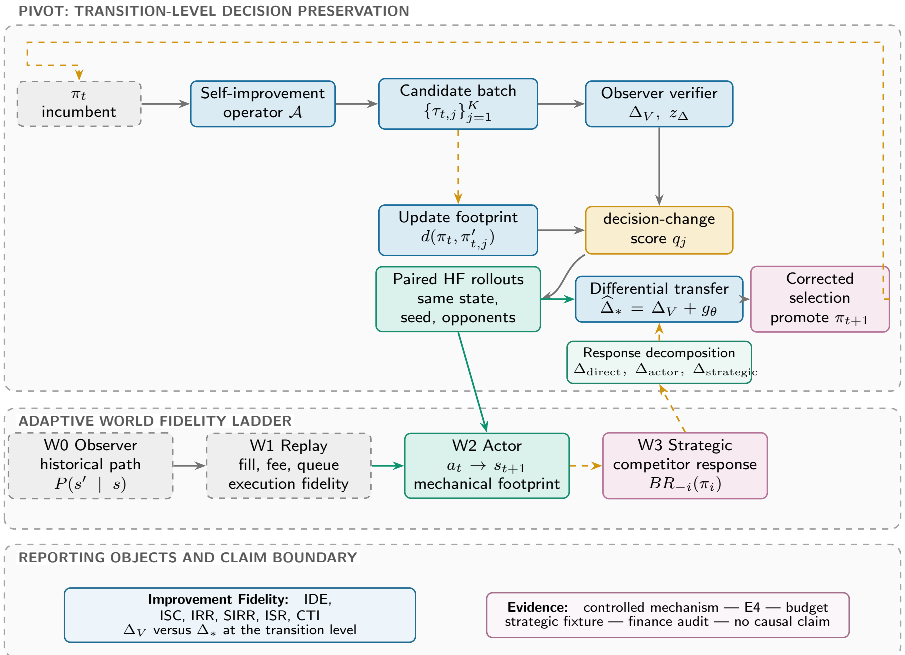

Figure 3: PIVOT validates the replacement operation rather than isolated policy values. A cheap verifier produces proxy improvements and update footprints; a decision-change score selects paired high-fidelity interventions under common state, seed, and opponent conditions. The differential correction ranks candidates for promotion and feeds the next self-improvement round.

Global versus differential fidelity. The matched-budget E4 comparison trains a global value model and a transition differential model on the same number of high-fidelity transition queries. Local-minus-global ISC is 0.1204; local-minus-global IDE is -1.7536. Update-selection regret is tied in this particular fixture, so we do not claim that a differential model dominates every policy-value baseline.

Value Fidelity versus Improvement Fidelity. We add a controlled evaluator contrast to isolate the estimand. Evaluator A has slightly lower isolated policy-value MAE (0.5489 versus 0.6000) and near-perfect policy ranking (0.9781 versus 1.0000), but its candidate-side bias grows with response and footprint. Evaluator B has a policy-independent value offset that is worse for isolated values but cancels in paired deltas. On the same 216 held-out transitions, A has IDE 1.0978, ISC 0.9444, IRR 0.1714, and ISR 0.2548, whereas B has IDE 0, ISC 1, IRR 0, ISR 0, and CTI -45.7229. A's CTI is -61.0081 (higher is better). This is a constructive controlled contrast, not a learned-model claim: similar policy-value accuracy does not determine local transition fidelity.

Budget frontier. At one high-fidelity query, PIVOT reduces ISR relative to Random by 4.7245 and relative to Top Proxy by 3.1301 in the registered fixture. fig. 5 shows the full frontier. The twelve-ablation suite retains a non-monotonic PIVOT curve across budgets; this is a robustness boundary, not a claim of monotonic benefit.

A required null. The E3 closed-loop fixture does not show proxy/true divergence: both curves increase together and reach the reward bound. We therefore report E3 as a negative stress test of the current fixture, not as evidence for "optimizing the wrong world". This distinction is important because a paper can otherwise manufacture its central phenomenon by selecting a pathological environment after observing results.

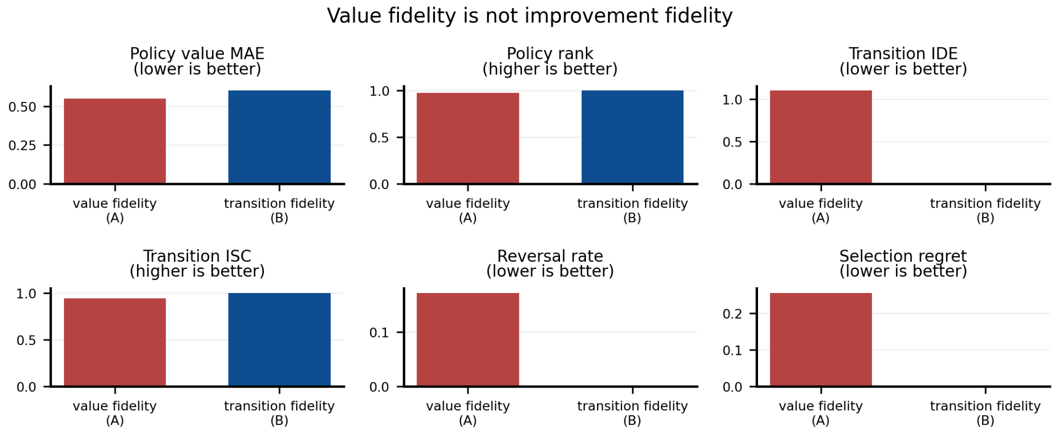

Figure 4: Policy-value and transition-differential predictions under matched high-fidelity budget. The estimands are related but not interchangeable.

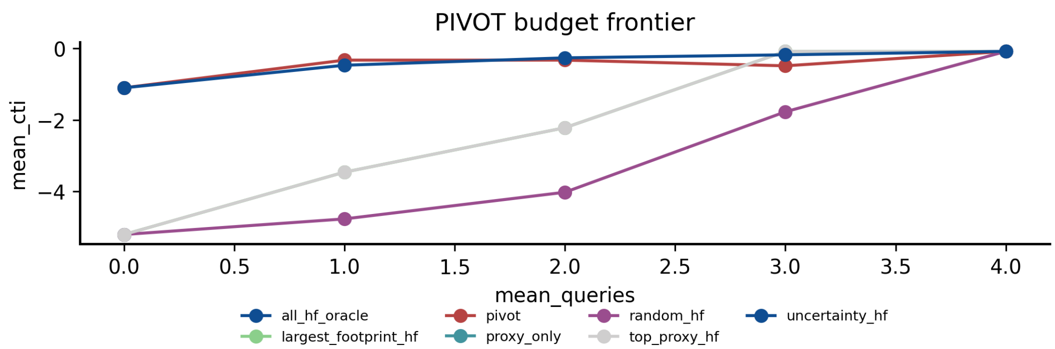

Figure 5: Matched-budget selection frontier. PIVOT spends queries on decision-sensitive transitions; All-HF is an oracle reference.

7 Stress Tests Beyond Controlled Environments

Pinned external adaptive development. We use version-pinned Farama MPE2 Farama Foundation (2025) through PettingZoo Terry et al. (2021). The short direct-replay observer and full actor rollout use paired, trajectory-disjoint seeds. Powered E3b and E4b comparisons do not support generic external CTI or transition-over-global advantages (E3b difference -3.8672), so neither supports an external closed-loop claim.

Strategic response. Only the focal policy self-improves. In the powered held-out finite-action best-response family, 140 opponent-seed clusters exceed the required 135 and the cluster strategic effect is -0.5884 (threshold -0.5). This supports strategic reversal for this opponent family only; no equilibrium or realistic-ecology claim is made. Diagnostics are in Appendix figs. 6 and 8.

Finance audit. The finance ladder is F0 historical path, F1 execution replay, F2 virtual fills, and F4 strategic response. Across 12 BTCUSDT, ETHUSDT, and BNBUSDT sessions, the public-data depth proxy has reversal 0/7 (primary) and 0/5 (holdout), with pooled effect -4.1554×10^{-7}. It is a plausibility/boundary test, not causal confirmation.

8 Ablations and Robustness

The frozen suite contains twelve comparisons; table 1 shows the compact subset and the snapshot contains full rows and failure ledgers.

Table 1: Selected controlled ablations. Values are descriptive estimates over three independent registered runs; slash-separated values compare variants.

Ablation	Estimate(s)	Metric
Paired vs unpaired	0.0433 / 0.0676	paired SE / unpaired SE
Footprint vs none	2.0668 / 2.0669	IDE; null
Small vs large	0.50 / 0.75	IRR
Weak vs strong response	0.458 / 1.000	IRR
Fixed vs adaptive	0.000 / 1.000	SIRR
Single vs multiple model	4.751 / 2.691	IDE
Candidate count 2 / 4	0.2368 / 0.2439	ISR
PIVOT budget 0 / 4	1.0210 / 0.0000	ISR

Pairing reduces standard error (0.0433 versus 0.0676); footprint, candidate, and budget variants are tied or non-monotonic. PIVOT is therefore a budgeted transition-validation mechanism, not a claim that every feature or budget increase helps.

9 Limitations and Claim Boundary

The controlled world establishes falsifiability rather than realism. Finance is observational with virtual fills and no live or private data. E3b/E4b are powered nulls; E7b is bounded to one held-out opponent family. LLM/EvoQuant and M3 adapters are deferred. We do not certify actions, solve equilibrium, or call a learned simulator ground truth.

10 Reproducibility and Conclusion

The snapshot contains 33 hash-indexed artifacts, seven figure/source-table pairs, an editable OpenTikZ bundle, evaluator-contrast rows, gate summaries, the public expansion, and the twelve-ablation aggregate. Three disjoint ablation runs reproduce with zero failures; tests, static checks, source commit, and PDF metadata are recorded in the release manifest. No credentials or raw vendor archives enter the package.

The lesson is that self-improvement must be judged by whether an update remains better in the world—and among the agents—that deployment causes to exist. PIVOT implements paired, transition-level, budgeted interventional validation; controlled evidence supports the estimand, while powered external tests bound generic CTI gains and expose a family-specific strategic reversal.

AI Use Statement. AI tools provided limited assistance with language editing and code/figure formatting; the authors verified all claims, computations, and artifacts and take full responsibility.

Reproducibility Statement. The anonymous supplement contains source, deterministic configurations, hash-indexed snapshot, table/figure inputs, and rebuild commands. The public-data audit uses virtual fills only and includes no credentials, private data, raw vendor archives, or live orders.

Ethics Statement. Public market data and controlled virtual environments only; no human subjects, personal data, live trading, or external authorization.

REFERENCES

Debabrota Basu, Udvas Das, Brahim Driss, and Uddalak Mukherjee. Performative policy gradient: Optimality in performative reinforcement learning, 2025. URL `https://arxiv.org/abs/2512.20576`.

David Byrd, Maria Hybinette, and Tucker Hybinette Balch. Abides: Towards high-fidelity market simulation for ai research. *arXiv preprint arXiv:1904.12066*, 2019. URL `https://arxiv.org/abs/1904.12066`.

Patrick Cheridito, Jean-Loup Dupret, and Zhexin Wu. Abides-marl: A multi-agent reinforcement learning environment for optimal execution with endogenous liquidity, 2025. URL `https://arxiv.org/abs/2511.02016`.

Rama Cont, Arseniy Kukanov, and Sasha Stoikov. Price dynamics in a markovian limit order market. *SIAM Journal on Financial Mathematics*, 1(1):1–29, 2010. doi: 10.1137/090752451. URL `https://arxiv.org/abs/1104.4596`.

Christoph Dann, Yishay Mansour, and Mehryar Mohri. Theoretical foundations and effective algorithms for policy-aware simulator learning, 2026. URL `https://arxiv.org/abs/2605.29032`.

Bo Deng, Kang Zhou, Lifan Guo, Chongyang Tao, Xuanren Chen, Chenggang Xie, Renzhao Liang, Feng Chen, and Chi Zhang. Finevo-bench: A longitudinal benchmark for self-evolving agents in professional financial workflows. *arXiv preprint arXiv:2608.06144*, 2026. URL `https://arxiv.org/abs/2608.06144`.

Farama Foundation. Mpe2: Multi-agent particle environment 2, 2025. URL `https://github.com/Farama-Foundation/MPE2`. Version 1.1.0, accessed 2026-08-25.

Diandian Guo, Cong Cao, Fangfang Yuan, Yingqi Wang, Yueshan Wang, and Dakui Wang. Self-authored verification is unreliable in heuristic self-improving agents, 2026. URL `https://arxiv.org/abs/2607.24300`.

Nicklas Hansen, Hao Su, and Xiaolong Wang. Td-mpc2: Scalable, robust world models for continuous control. In *International Conference on Learning Representations*, 2024. URL `https://arxiv.org/abs/2310.16828`.

Carlos E. Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik Narasimhan. Swe-bench: Can language models resolve real-world github issues? In *International Conference on Learning Representations*, 2024. URL `https://arxiv.org/abs/2310.06770`.

Maxime Kawawa-Beaudan, Srijan Sood, Kassiani Papasotiriou, Daniel Borrajo, and Manuela Veloso. Tradefm: A generative foundation model for trade-flow and market microstructure. *arXiv preprint arXiv:2602.23784*, 2026. URL `https://arxiv.org/abs/2602.23784`.

Vivek Kulkarni, Sudipta Paul, Aounon Kumar, Nicholas Tzou, and Srinivas Chappidi. Sbco: Self-supervised, verifier-grounded harness optimization for planning agents. *arXiv preprint arXiv:2608.10157*, 2026. URL `https://arxiv.org/abs/2608.10157`.

Jialong Li and Jialing Zhu. Auditing self-evolution in financial agents: Capability gains, security drift, and execution-interface mismatch. *arXiv preprint arXiv:2608.17684*, 2026. URL `https://arxiv.org/abs/2608.17684`.

Dennis V. Lindley. On a measure of the information provided by an experiment. *The Annals of Mathematical Statistics*, 27(4):986–1005, 1956. doi: 10.1214/aoms/1177728069.

Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, Minlie Huang, Yuxiao Dong, and Jie Tang. Agentbench: Evaluating llms as agents. In *International Conference on Learning Representations*, 2024. URL `https://arxiv.org/abs/2308.03688`.

Xiao-Yang Liu, Ziyi Xia, Jingyang Rui, Jiechao Gao, Hongyang Yang, Ming Zhu, Christina Dan Wang, Zhaoran Wang, and Jian Guo. Finrl-meta: Market environments and benchmarks for data-driven financial reinforcement learning. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, volume 35, 2022. URL https://arxiv.org/abs/2211.03107.

Debmalya Mandal, Stelios Triantafyllou, and Goran Radanovic. Performative reinforcement learning. *arXiv preprint arXiv:2207.00046*, 2022. URL https://arxiv.org/abs/2207.00046.

Jie Mao, Changlun Li, Xiang Li, Qiqi Duan, Jinhui Yuan, Xiang Liu, Yuyu Luo, Jing Tang, Xiaowen Chu, and Nan Tang. Evoquant: Self-evolving verifier-guided strategy optimization for robust quantitative trading, 2026. URL https://arxiv.org/abs/2607.12455.

Yuriy Nevmyvaka, Yi Feng, and Michael Kearns. Reinforcement learning for optimized trade execution. In *Proceedings of the 23rd International Conference on Machine Learning*, pp. 673–680, 2006. doi: 10.1145/1143844.1143929.

Patrick Noble, Mathieu Rosenbaum, and Saad Souilmi. Bridging the reality gap in limit order book simulation. *arXiv preprint arXiv:2603.24137*, 2026. URL https://arxiv.org/abs/2603.24137.

Juan C. Perdomo, Tijana Zrnic, Celestine Mendler-Dunner, and Moritz Hardt. Performative prediction. *arXiv preprint arXiv:2002.06673*, 2020. URL https://arxiv.org/abs/2002.06673.

Georgios Piliouras and Fang-Yi Yu. Multi-agent performative prediction: From global stability and optimality to chaos. *arXiv preprint arXiv:2201.10483*, 2022. URL https://arxiv.org/abs/2201.10483.

Julian Quevedo, Ansh Kumar Sharma, Yixiang Sun, Varad Suryavanshi, Percy Liang, and Sherry Yang. Worldgym: World model as an environment for policy evaluation, 2025. URL https://arxiv.org/abs/2506.00613.

Rilind Sahitaj, Paulius Sasnauskas, Yigit Yalin, Debmalya Mandal, and Goran Radanovic. Independent learning in performative markov potential games. *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics*, 2025. URL https://arxiv.org/abs/2504.20593.

Burr Settles. Active learning literature survey. Technical Report 1375, University of Wisconsin–Madison, 2009. URL https://minds.wisconsin.edu/handle/1793/60660.

Justin K. Terry, Benjamin Black, Nathaniel Grammel, Ananth Jayakumar, Ananth Hari, Ryan Sullivan, Luis Santos, Clemens Perez, Christopher Horsch, Prasoon Ravi, et al. Pettingzoo: Gym for multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 34, 2021. URL https://arxiv.org/abs/2009.14471.

Guanghui Wang, Krishna Acharya, Lokranjan Lakshmikanthan, Juba Ziani, and Vidya Muthukumar. Last-iterate convergence for symmetric, general-sum, 2×2 games under the exponential weights dynamic, 2025. URL https://arxiv.org/abs/2502.08063. Includes a multi-agent performative prediction application.

Zhilin Wang, Han Song, Runzhe Zhan, Jusen Du, Jiacheng Chen, Tianle Li, Qingyu Yin, Yulun Wu, Zhennan Shen, Tong Zhu, Yanshu Li, Guanjie Chen, Derek F. Wong, Yafu Li, Yu Cheng, and Yang Yang. Evopolicygym: Evaluating autonomous policy evolution in interactive environments, 2026. URL https://arxiv.org/abs/2607.02440.

Tianxin Wei, Zhan Shi, Minhua Lin, Bing He, Zewen Liu, Yisi Sang, Yuanchen Bei, Xuying Ning, Jiaru Zou, Ting-Wei Li, Xiao Lin, Yanjun Zhao, Chi Wang, Benoit Dumoulin, Dakuo Wang, Jingrui He, and Hanqing Lu. Evo-harness: Context-to-harness skill compilation for self-evolving agents. *arXiv preprint arXiv:2608.15071*, 2026. URL https://arxiv.org/abs/2608.15071.

Qinyuan Ye, Yu Li, Yada Pruksachatkun, Jiaxin Zhang, and Chien-Sheng Wu. On the fragility of self-improving agents: Variance, task order, and underspecification. *arXiv preprint arXiv:2608.18066*, 2026. URL https://arxiv.org/abs/2608.18066.

Jenny Zhang, Shengran Hu, Cong Lu, Robert Lange, and Jeff Clune. Darwin godel machine: Open-ended evolution of self-improving agents. *arXiv preprint arXiv:2505.22954*, 2025. URL https://arxiv.org/abs/2505.22954.

Yanzhi Zhang, Yu Ma, Yilin Cheng, Jian Li, and Yitong Duan. M3: A state-event generative foundation model for market microstructure dynamics. *arXiv preprint*, 2026. URL https://arthurzhang02.github.io/m3-market-microstructure/M3_paper.pdf.

Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. Webarena: A realistic web environment for building autonomous agents. In *International Conference on Learning Representations*, 2024. URL https://arxiv.org/abs/2307.13854.

Mingxi Zou, Jiaxiang Chen, Aotian Luo, Jingyi Dai, Chi Zhang, Dongning Sun, and Zenglin Xu. Finevo: From isolated backtests to ecological market games for multi-agent financial strategy evolution. *arXiv preprint arXiv:2602.00948*, 2026. URL https://arxiv.org/abs/2602.00948.

A Artifact and Full Result Notes

The frozen artifact manifest is `../snapshot/manifest.json`. Its controlled summary hashes are inherited from the clean-room reproduction. The ablation aggregate is identified by the manifest hash prefix `e081f87f`; the public expansion remains marked `causal_impact_identified=false` and `live_orders=false`. Re-running the snapshot and table scripts from a new output directory is required before changing any result table.

Table 2: Controlled and public evidence summary.

Experiment	Estimate	Interpretation
P2 reversal	0.9722	IRR, high response
E4 local–global	0.1204	ISC difference
E5 Random–PIVOT	4.7245	ISR at HF budget 1
E5 Top Proxy–PIVOT	3.1301	ISR at HF budget 1
E6 zero participation	0.0000	F2–F1
E7 SIRR	1.0000	strategic reversal
E8 SIRR	1.0000	strategic reversal
E9 CTI	-1.5338	8-round exploratory loop

Table 3: Public expansion boundary.

Public audit field	Value
Asset/date sessions	12
F1-positive sessions	7
Primary reversals	0/7
Holdout reversals	0/5
Pooled depth effect	-4.1554×10^{-7}
Causal impact identified	No

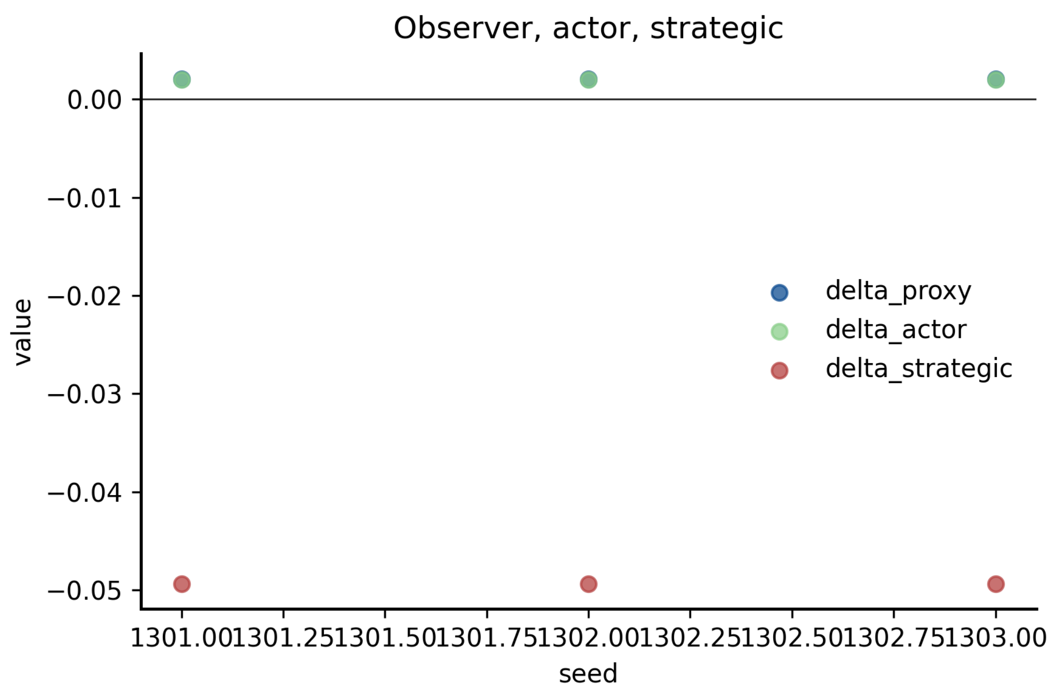

Figure 6: The same update evaluated in observer, actor, and strategic worlds. This diagnostic separates mechanical response from competitor adaptation; it is fixture evidence rather than a general-equilibrium claim.

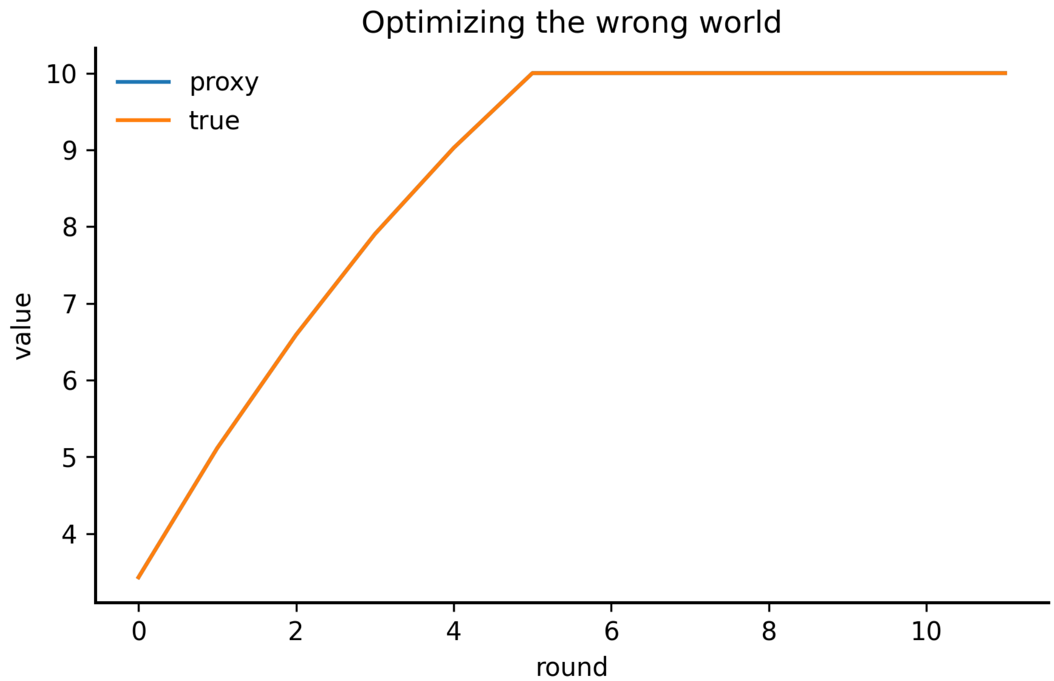

Figure 7: Required E3 null: proxy and true values overlap and saturate at the reward bound. The current closed-loop fixture therefore does not establish performative overoptimization.

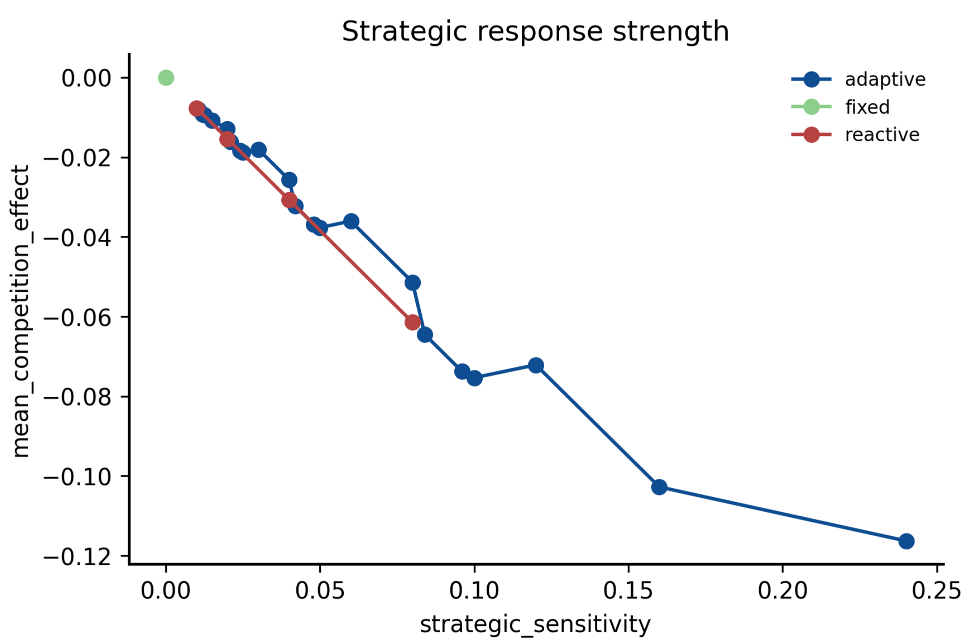

Figure 8: Strategic response effect across opponent modes and sensitivities. This appendix diagnostic is fixture evidence, not a general-equilibrium claim.